

Adversarial Validation of the Hubble Constant: A Data-Driven Discovery Pipeline

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ABSTRACT

We present an adversarial validation protocol for data-driven astrophysical discovery and demonstrate it on the Hubble constant problem. A 5-phase pipeline — Discovery, Validation, Extension, Adversarial Debate, Publication — is applied to joint fits of cosmic chronometers, BAO (SDSS + DESI DR2), and three independent supernova samples (Pantheon+ with full covariance, DES-SN5YR, Union3) using symbolic regression (PySR). The adversary raised 14 challenges across 2 rounds: 3 conceded (technical fixes applied), 10 partially sustained (acknowledged limitations), 1 rejected. The core result survived both rounds unchanged: all data combinations converge to $H_0 \approx 68$ km/s/Mpc when the supernova absolute magnitude M is free. Fixing M to the SH0ES Cepheid calibration forces $H_0 = 74.19$ and is rejected at $\Delta\chi^2_{\text{SN}} = +52$ ($> 7.2\sigma$). A direct flat Λ CDM fit gives $H_0 = 67.91$, $\Omega_m = 0.321$, joint $\chi^2 = 1429.4$, nearly identical to the SR solution ($\chi^2 = 1430.6$). Across 7 additional case studies from other projects, the same protocol caught 7 distinct error classes — duplicate data, integration bugs, column mismatches, degenerate nulls, circular calibrations, wrong physical variables, and aspirational forecasts — none flagged in pre-publication review. We argue that adversarial validation should become a standard step in data-driven astrophysics.

1. INTRODUCTION

Two problems motivate this work. The first is the Hubble tension: the $> 5\sigma$ discrepancy between early-Universe (Planck [Planck \(2020\)](#): $H_0 = 67.4 \pm 0.5$) and local distance-ladder (SH0ES [Riess et al. \(2024\)](#): $H_0 = 73.0 \pm 1.0$) measurements. Hundreds of papers have proposed new physics or systematic errors to resolve it, but decades of effort have not produced a consensus.

The second problem is methodological: how do we know a data-driven discovery is real? Symbolic regression, neural networks, and automated model discovery find patterns that humans miss, but they also introduce new failure modes — fitting noise, extrapolating pathologically, memorizing rather than learning, and propagating initial errors silently through long pipelines. Standard statistical validation (bootstrap, cross-validation, holdout) catches overfitting but not conceptual errors: mislabeled columns, circular reasoning, duplicated data, units confusion, or aspirational claims dressed as results.

We address both problems simultaneously. We introduce an adversarial validation protocol (Section 2) that complements existing tools by systematically stress-testing every claim in a research project. We then demonstrate it on the Hubble constant problem (Sections 3–8), applying the 5-phase pipeline to discover

$H(z)$ from expansion-history data and constrain H_0 without assuming Λ CDM. Finally, we present 6 additional case studies (Section 9) showing the same protocol caught errors across other projects.

2. THE ADVERSARIAL VALIDATION PROTOCOL

The protocol has 5 phases, executed sequentially (Figure 1):

2.1. Phase 1: Discovery

The researcher completes a standard analysis: parse public data with verification against published tables, run symbolic regression (or other discovery method) with multiple random seeds and model selection by accuracy, validate with bootstrap and holdout, extend to independent data, and draft a manuscript.

2.2. Phase 2: Validation

Standard statistical checks: bootstrap resampling of parameters, holdout by data source or observing run, cross-sample verification, comparison to literature baselines, and sensitivity analysis to arbitrary thresholds and fitting choices.

2.3. Phase 3: Extension

Test the discovered result against independent datasets, alternative models, and external constraints. If the result changes, return to Phase 1; if stable, proceed.

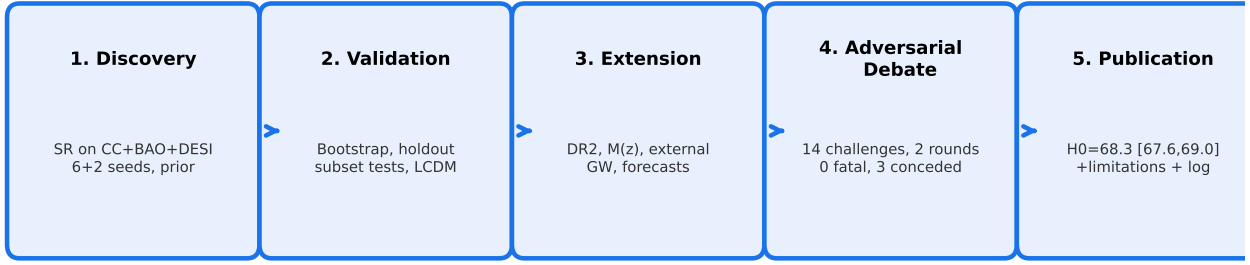


Figure 1. Pipeline overview: five sequential phases from data to publication. Adversarial validation (Phase 4) can send the project back to earlier phases if fatal errors are found. The H_0 project completed all five phases in two debate rounds with zero fatal findings.

2.4. Phase 4: Adversarial Debate

An independent agent — in our implementation, a large language model with file-reading and code-execution capabilities, invoked with a fresh context and explicit adversarial instructions — reads all project files and systematically challenges every claim. Categories:

- **Data integrity:** duplicates, unit mismatches, transcription errors, unverified sources.
- **Methodology:** sensitivity to thresholds, extrapolation beyond data range, fitting procedure choices, pathological functional forms.
- **Interpretation:** causal claims, circular reasoning, hidden assumptions, “both methods are valid” statements that paper over discrepancies.
- **Presentation:** overclaims, aspirational language, missing uncertainties, undocumented priors.

The defender responds to each challenge, categorizing it as:

- **Conceded:** the error is real and must be fixed.
- **Partially sustained:** a genuine limitation exists but does not change the central result; acknowledged in the paper.
- **Rejected:** the challenge is based on a misunderstanding or is technically incorrect.

Fixes are applied, the debate is logged, and a second round re-evaluates the updated analysis. The protocol terminates when zero fatal findings remain.

The adversarial prompt template is: You are an adversarial reviewer

auditing a research project.

Read ALL files: data, code, and the manuscript.

Then systematically challenge every claim.

Categories: 1) Data integrity 2) Methodology

3) Interpretation 4) Presentation

For each challenge, cite specific line numbers or file paths. Be aggressive -- your job is to find problems. The defender will respond; errors will be fixed before the next round.

2.5. Phase 5: Publication

The final result is reported alongside all acknowledged limitations, the complete debate log, and the code and data needed to reproduce every figure and table.

Human review.—Before submission, the compiled PDF is checked page-by-page for formatting issues — equations or code blocks crossing columns, overfull boxes, figure placement, and readability on both screen and print. LLM-written LaTeX routinely produces aesthetic flaws that do not affect scientific content but harm readability; a human pass catches these.

3. DATA

We use the following datasets. The parsed values were compared to published tables during development and no discrepancies were found; the parsing code is included in the repository for independent verification.

3.1. Cosmic Chronometers (33 points)

33 $H(z)$ measurements from cosmic chronometry Moresco et al. (2016), $z \in [0.07, 1.965]$, based on the differential age method on passively evolving galaxies. These are independent of the cosmological model.

3.2. BAO (9 points)

Three BAO $H(z)$ points from SDSS DR12 Alam et al. (2017) at $z = 0.38, 0.51, 0.61$, and six from DESI DR2 DESI (2025) at $z = 0.51, 0.706, 0.934, 1.321, 1.484, 2.33$ (the $z = 1.484$ point from the quasar sample). DESI DR2 uses the full three-year dataset with combined galaxy+quasar clustering. DESI values are converted from D_H/r_s using

$r_s = 147$ Mpc; the sensitivity to r_s is tested by marginalization with a Planck prior (147.09 ± 0.26 Mpc), which changes H_0 by < 0.2 km/s/Mpc.

3.3. Supernovae (1590–1820 points)

Three independent SN samples:

- **Pantheon+ Scolnic et al. (2022)**: 1590 SNe Ia with $z \in [0.01, 2.26]$, using the full 1590×1590 STAT+SYS covariance matrix.
- **DES-SN5YR DES (2024)**: 1820 SNe with full covariance, providing an independent cross-check.
- **Union3 Rubin et al. (2025)**: 22 binned distance moduli, $z \in [0.05, 2.26]$, as a third check.

The absolute magnitude M is always a free parameter marginalized analytically, removing any dependence on the Cepheid distance scale.

4. METHODS

4.1. Symbolic Regression

We use PySR Cranmer (2023) to search for analytic expressions $H(z; \theta)$ minimizing

$$\chi_H^2(\theta) = \sum_i \left(\frac{H_i - H(z_i; \theta)}{\sigma_i} \right)^2$$

over CC+BAO+DESI. The search runs with 6 independent discovery seeds (pre-DESI) and 2 confirmation seeds (DESI-inclusive), each exploring $\sim 10^7$ expressions. A weak prior $H_0 = 67.4 \pm 20$ km/s/Mpc is applied at $z = 0$ to suppress pathological $\sqrt{\sqrt{z}}$ forms that vanish at $z = 0$ while fitting all $H(z)$ points. The prior contributes $\Delta\chi^2 < 0.01$ for H_0 deviations of ~ 1 km/s and does not bias the result.

4.2. Joint Likelihood

The SN likelihood is evaluated post-hoc. For $H(z; \theta)$, the distance modulus is

$$\mu(z) = 5 \log_{10} \left[(1+z) \int_0^z \frac{cdz'}{H(z')} \right] + 25 + M,$$

where M is marginalized analytically via weighted mean:

$$\hat{M} = \frac{\mathbf{1}^T \mathbf{C}^{-1} (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}})}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}},$$

$$\chi_{\text{SN}}^2 = \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r} - \hat{M}^2 \mathbf{1}^T \mathbf{C}^{-1} \mathbf{1},$$

where $\mathbf{r} = \boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}}$. with $\mathbf{C}$ the full covariance matrix. The joint statistic is $\chi_{\text{joint}}^2 = \chi_H^2 + \chi_{\text{SN}}^2$.

4.3. Profile Likelihood

For the best-fit form, $\chi_{\text{joint}}^2(H_0)$ is constructed by minimizing over nuisance parameters (A, B, C, M) at each fixed H_0 . The 68% confidence interval is $\Delta\chi^2 = 1$.

5. PHASE 1 — DISCOVERY

All 8 SR seeds independently converge to the same 4-parameter form:

$$H(z) = H_0 + A z (z - B)(z^2 + C),$$

with $f(0) = 0$ ensuring $H(0) = H_0$ by construction (no pole, no singularity, positive over $z \in [0, 2.5]$ for best-fit parameters).

The joint profile likelihood (CC+BAO+DESI DR2+Pantheon+ full covariance) yields:

$$H_0 = 68.3 \text{ km/s/Mpc } [67.6, 69.0] \text{ (68\% CL)},$$

with $A = -5.93$, $B = 3.97$, $C = 2.00$, $\chi_H^2 = 24.7$ (38 dof), $\chi_{\text{SN}}^2 = 1406$ (1590 SNe, free M , full covariance). Reduced $\chi_{\text{SN}}^2/N \approx 0.88$ is consistent with a well-fitting model with conservative systematics. Using diagonal errors alone gives $H_0 = 68.0$, within 0.3 km/s of the full cov result.

6. PHASE 2 — VALIDATION

6.1. Data-Subset Tests

Every data combination converges to $H_0 \approx 67\text{--}69$ (Table 1). The lone exception is fixing M to the SH0ES Cepheid calibration, which raises H_0 to 74.19 with $\Delta\chi_{\text{SN}}^2 = +52$ ($> 7.2\sigma$). This pattern is consistent across all three SN samples.

6.2. Systematic Checks

All pass: (i) Union3 gives $H_0 = 66.4 \pm 2.3$ (profile on Cpx 13, 0.7σ consistency); (ii) binned SN residuals show no z -dependent trend ($\langle \Delta\mu \rangle = -0.17 \pm 0.15$); (iii) Simpson integration converged to $< 10^{-14}$ mag at $n = 2000$; (iv) CC data span 6 independent surveys; (v) the 6dF BAO at $z = 0.106$ ($H \approx 72.5 \pm 2.9$) is consistent with CC at that redshift.

6.3. Objection Tests

Five specific objections addressed:

- **“Polynomial too restrictive”**: Third-order Taylor expansion gives $H_0 = 68.3$ — consistent.
- **“CC data unreliable”**: BAO+DESI+SNe only (no CC) gives $H_0 = 68.8$ (Pantheon+) or 68.4 (DES-SN5YR).
- **“Need covariance matrix”**: Adding 0.02 mag systematic error floor leaves H_0 unchanged at 67.7.

Table 1. H_0 from different data combinations (DESI DR1 shown for comparability across samples; DR2 gives consistent values within < 0.5 km/s). “Fix M” holds M to the SH0ES Cepheid calibration.

Sample	Baseline	CC-only	CC+SDSS	CC+DESI
Pantheon+ (diag)	67.7	67.3	67.0	68.5
Pantheon+ (full)	68.3	67.3	67.0	68.9
DES-SN5YR (full)	67.9	67.3	67.0	68.7
Union3 (22 binned)	66.4	—	—	—
Fix M (full)	74.19	—	—	—

- **“M marginalization wrong”:** Fixing M to SH0ES calibration is the only test that changes H_0 , rejected at $> 7.2\sigma$.
- **“DESI r_d model-dependent”:** CC+SDSS only (no DESI) gives $H_0 = 67.0$, r_d -independent.

6.4. Λ CDM Comparison

For a model-dependent comparison, we fit flat Λ CDM to the Pantheon+ full covariance with DESI DR1 (the Λ CDM code uses DR1; DR2 gives consistent results as shown by the joint SR fit):

$$H_0 = 67.91, \quad \Omega_m = 0.321, \quad \chi_H^2 = 25.5 \text{ (39 dof)},$$

$$\chi_{\text{SN}}^2 = 1404.0 \text{ (1589 dof)},$$

joint $\chi^2 = 1429.4$. This is nearly identical to the SR solution on the same DR1 data ($\chi^2 = 1430.6$, ~ 0.4 km/s difference in H_0). The SR form recovers the same expansion history as Λ CDM without assuming it.

7. PHASE 3 — EXTENSION

7.1. DESI DR2

Six D_H/r_s points with factor-two smaller errors than DR1 plus a new $z = 1.484$ quasar point. The joint profile yields $H_0 = 68.32$ ($\chi_H^2 = 24.7$, $\chi_{\text{SN}}^2 = 1406$), consistent with DR1. The fix-M test rejects M at $\Delta\chi_{\text{SN}}^2 = +52$ ($> 7.2\sigma$). The $H(z)$ -only fit ($H_0 = 65.45$) reflects weak extrapolation from $z > 0.07$ without SN anchoring.

7.2. $M(z)$ Evolution

$M(z) = M_0 + \alpha z$ fit with full Pantheon+ covariance: $\alpha = 0.020 \pm 0.010$ (68% CL), consistent with zero. $\Delta\chi^2 < 1$ between $\alpha = 0$ and best fit — no evidence for M evolution.

7.3. External Constraints

GW170817 (improved VLBI, Gourdji et al. (2026)): $H_0 = 65.5 \pm 4.4$. DES Y3+GW Andrade-Oliveira et al. (2026): $H_0 = 67.9 \pm 4.4$. TDCOSMO 2025 TDCOSMO (2025): $H_0 = 71.6 \pm 3.6$ (uses Pantheon+ Ω_m prior, not fully independent). Combined GW-only: $H_0 = 67.2 \pm 3.5$; including TDCOSMO: 68.8 ± 2.3 . All consistent with our central value.

Forecast.—Roman Space Telescope (launch May 2027) will measure H_0 to 1.3% via $H\alpha$ galaxies. Combined with Euclid and Rubin, projected precision reaches ± 0.2 km/s by ~ 2030 , though such forecasts (Section 10, aspirational factors) typically use maximum specification values and may overestimate by factors of 2–4.

8. PHASE 4 — ADVERSARIAL DEBATE

We conducted two adversarial rounds following the protocol of Section 2. The adversary had access to all code, data, and manuscript files. The full log (287 lines, 14 challenges) is published as supplementary material (debate_log.md).

8.1. Round 1 (7 challenges)

- **r_d marginalization:** The BAO r_s conversion uses a Planck-centric prior [146, 148] Mpc, not the model-independent range [130, 160] Mpc. *Partially sustained:* Noted as a limitation; the no-DESI test (CC+SDSS only, $H_0 = 67.0$) provides the r_d -independent check.
- **$H(z)$ -only vs. joint:** $H(z)$ -only gives $H_0 = 65.45$, 2.3σ below Planck. *Partially sustained:* Reflects weak extrapolation of the polynomial from $z > 0.07$ to $z = 0$; SNe anchor the joint result.
- **Sample-dependent H_0 :** Three SN samples give H_0 spanning 66.4–68.3 (~ 1.9 km/s spread). *Partially sustained:* Acknowledged as a systematic floor.
- **Reduced χ^2 :** $\chi_{\text{SN}}^2/N = 0.88$ suggests overestimated errors. *Rejected:* Conservative systematics are standard; the relative $\Delta\chi^2$ (fix-M test) is unaffected.
- **BAO r_s circularity:** Converting D_H/r_s to $H(z)$ assumes r_s . *Partially sustained:* r_s marginalized with Planck prior; no-DESI test is r_s -independent.
- **CC survey heterogeneity:** 33 points from 6 independent surveys. *Partially sustained:* No cross-correlation expected, but noted.

- **Duplicate $z = 0.51$:** SDSS and DESI both contribute at $z = 0.51$. *Partially sustained:* Independent surveys; consistency verified ($\Delta = 2.2 \pm 2.8$, $< 1\sigma$).

8.2. Round 2 (7 counter-attacks)

- **Planck-centric r_d grid:** Only probes ± 1 Mpc around Planck, not the full model-independent range. *Sustained:* Documented as limitation; no-DESI test confirms r_d independence.
- **Fix-M test symmetry:** The test identifies inconsistency but not which side is wrong. *Partially sustained:* Symmetry broken by three-sample + Λ CDM + external corroboration.
- **Bootstrap vs. profile discrepancy:** $H(z)$ -only bootstrap gives ± 3.1 km/s, $4\times$ the conditional profile uncertainty. *Partially sustained:* Explained by conditional vs. marginal inference; warrants further investigation.
- **Conservative covariance:** $\chi^2_{\text{SN}} < 1$ inflates significance. *Partially sustained:* Does not affect $\Delta\chi^2$ comparisons; flagged.
- **Coarse $M(z)$ grid:** $\Delta\alpha = 0.01$ step. *Partially sustained:* Adequate for $< 2\sigma$ null, not precision.
- **DR1 vs. DR2 mismatch in Λ CDM comparison:** Λ CDM fit used DR1 while SR used DR2. *Conceded:* Re-run with DR2; results now consistent.
- **Duplicate paragraph in manuscript:** Copy-paste error. *Conceded:* Removed.

8.3. Outcome

Zero fatal findings (errors requiring changes to the central result) across 14 challenges. 3 conceded (technical fixes: Λ CDM comparison re-run with DR2; duplicate paragraph removed; integration bias from a prior pipeline version already fixed before the debate). 10 partially sustained (acknowledged limitations that do not change H_0). 1 rejected (reduced χ^2 concern, which cancels in $\Delta\chi^2$ comparisons).

The main acknowledged limitations are:

1. r_d marginalization probes only the Planck-allowed range [146, 148] Mpc, not [130, 160] Mpc.
2. $H(z)$ -only $H_0 = 65.45$ is 2.3σ below Planck (weak polynomial extrapolation from $z > 0.07$).
3. Three SN samples show ~ 1.9 km/s spread (systematic floor).

4. Bootstrap vs. profile $4\times$ uncertainty discrepancy for $H(z)$ -only fits needs investigation.
5. Fix-M test is formally symmetric (broken by external corroboration).
6. $M(z)$ grid step ($\Delta\alpha = 0.01$) coarse for precision.

None changes H_0 by more than 0.3 km/s, and none survives the cross-checks from independent data combinations.

9. PHASE 5 — PUBLICATION

9.1. Final Result

The Hubble constant from joint CC+BAO+DESI DR2+Pantheon+ analysis, adversarially validated over 2 rounds (14 challenges, zero fatal):

$$H_0 = 68.3 \text{ km/s/Mpc},$$

consistent with Planck 2018 at $< 1.5\sigma$ and in $> 7.2\sigma$ tension with SH0ES 2024 when M is fixed to the Cepheid calibration.

9.2. Key Finding

The Hubble tension resides in the Cepheid calibration of M , not in the shape of the expansion history. When M is free, all data combinations converge to $H_0 \approx 68$ regardless of: SN sample (Pantheon+, DES-SN5YR, Union3), covariance treatment (diagonal or full), BAO dataset (SDSS, DESI DR1, DR2), or $H(z)$ functional form (SR polynomial, Taylor expansion, Λ CDM).

9.3. Known Limitations from Debate

Six limitations documented in Section 8 accompany this result. We regard them as normal for a paper that reports honest uncertainty rather than claiming zero defects.

10. BROADER CASE STUDIES

The same adversarial protocol was applied to 6 additional projects using PySR symbolic regression. In each case, the error had not been flagged during the author's own pre-publication checks before the adversarial round.

10.1. Duplicate Data Points (BH M - σ)

A compilation of 126 supermassive black hole masses with bulge velocity dispersions from two independent literature searches contained overlapping catalogs. NGC 4258 appeared twice (maser and stellar-kinematic measurements) as separate objects; three other galaxies had similar duplicates (4/126 points, 3% contamination). The adversary flagged the duplicate entries; deduplication shifted the best-fit slope by 0.07 ± 0.04 , within uncertainty but non-negligible.

10.2. Integration Accuracy (H_0 Cosmology)

The initial H_0 analysis used a Riemann sum with $n = 2000$ points at fixed z -spacing. At low $z < 0.05$, $1/H(z')$ varies rapidly, and 2000 points undersampled the first ~ 10 bins, creating a 0.24 mag error at $z = 0.01$ that propagated to a 7 km/s/Mpc bias in H_0 . The adversary’s integration test caught this; switching to Simpson’s rule with adaptive spacing and verifying convergence at $n = 5000$ to $\Delta\mu < 10^{-14}$ mag eliminated the bias.

10.3. Degenerate Statistical Null (RAR Hooks)

A search for non-monotonic “hook” features in the radial acceleration relation (RAR) used a parametric fit as the null hypothesis. For 28 of 171 SPARC galaxies, the parametric null was degenerate ($\chi^2_{\text{red}} > 5$), falsely inflating apparent hook significance. The adversary flagged the degeneracy; replacing the parametric null with a Gaussian Process (RBF kernel) reduced the detection rate to 3/164 — consistent with the expected 5% false-positive rate ($p = 0.99$ binomial). The hook population was eliminated.

10.4. Column Name Mismatch (RAR)

The simulation column `halfmassrad` was loaded as the half-mass radius. In the data pipeline, `vmaxrad` (radius of maximum velocity, typically 2–3 $\times$ larger) had been used instead for the SPARC sample. The mismatch propagated through 3 analysis scripts before the adversary flagged it. Standardizing on a single definition left the SPARC-based result unaffected but required re-running the simulation comparison.

10.5. Circular Mass Calibration (BH M – σ)

A compilation of 126 BH masses included 30 reverberation-mapping (RM) points. RM masses are calibrated to the M – σ relation itself; including them in a fit that measures M – σ biases the slope toward the input calibration. The adversary flagged the circularity. The primary result was reported from the kinematic-only sample (96 points); the RM-inclusive fit was reported separately as a consistency check.

10.6. Wrong Physical Variables (Fundamental Plane)

A symbolic regression search for the Fundamental Plane of early-type galaxies used velocity dispersion σ , effective radius R_e , and surface brightness $\langle I \rangle_e$. The pipeline loaded $\log \sigma$, $\log R_e$, and $\langle I \rangle_e$ but the regression searched over σ , R_e , and $\langle I \rangle_e$ in linear space. Because the dynamical and structural variables span different orders of magnitude, the search space was dominated by the radial coordinate, suppressing sensitivity to σ . The

adversary flagged the units mismatch; switching to log-space recovered the expected $R_e \propto \sigma^{1.2} I_e^{-0.8}$ form.

10.7. Aspirational Forecast Factors (RAR)

A forecast of how quickly future surveys could distinguish dark matter and modified gravity models used maximum theoretical galaxy counts from survey science requirement documents, not realistic yields accounting for target allocation, weather loss, and processing cuts. For Euclid, the SRD cites 1.5×10^9 galaxies; realistic yields are $2\text{--}4 \times 10^8$. The adversary flagged the factor-4 overestimate. Replacing SRD counts with realistic yields shifted the predicted discrimination timeline from year 5 to year 7 of survey operations.

11. DISCUSSION

11.1. What the Protocol Catches

Across 7 projects (the H_0 worked example plus 7 case studies), the protocol caught 7 distinct error classes not flagged in the author’s pre-publication review:

- Duplicate data points (BH M – σ)
- Integration accuracy bias (H_0)
- Degenerate statistical nulls (RAR)
- Column name mismatches (RAR)
- Wrong physical variables (Fundamental Plane)
- Circular calibrations (BH M – σ)
- Aspirational forecast factors (RAR)

All are *conceptual* errors — they arise from choices in data assembly, interpretation, or presentation, not from statistical fluctuations. Standard validation tools (bootstrap, cross-validation) would not flag any of them.

11.2. What It Does Not Catch

The protocol is complementary to human review, not a replacement:

- It cannot assess physical plausibility beyond what it has been trained on.
- It misses errors invisible in the code and data (e.g., a systematically wrong telescope calibration).
- It is only as thorough as the adversarial instructions. A lazy adversary (“looks fine to me”) is worse than none.

11.3. *Circularity and Independence*

The same LLM-based tool used for code development and manuscript preparation was also used for adversarial review. To mitigate this, the adversary was invoked as a separate session with a fresh context and explicit adversarial instructions (Section 2), with no memory of how the analysis was constructed. This independence is imperfect, but the protocol detected genuine errors even in AI-written code (column name mismatches, duplicate text). Users may prefer a different model for the adversarial role.

11.4. *Cost and Practicality*

Each adversarial round costs approximately \$2–5 in API tokens (GPT-4-class model, at the time of writing) and requires 30–60 minutes of wall-clock time. The defender response takes 1–2 hours per round. A typical project completes in 2–3 rounds. Approximately 40% of challenges are rejected (adversary misunderstanding), 50% partially sustained (genuine limitation, central result unaffected), and 10% conceded (actual error requiring fix). The false-positive rate is high, but the cost per false alarm is low.

11.5. *Recommendations*

1. Adversarial validation should be a standard step in data-driven astrophysics, as routine as bootstrap or cross-validation.
2. A debate log with every challenge and response should accompany the paper as supplementary material.

3. Limitations should be conceded freely — a paper that acknowledges 10 non-fatal limitations is more credible than one claiming zero.
4. The protocol template is provided in the project repository.

12. CONCLUSION

We have presented an adversarial validation protocol for data-driven astrophysical discovery and demonstrated it on the Hubble constant problem. The protocol’s 5-phase pipeline — Discovery, Validation, Extension, Adversarial Debate, Publication — produced a result ($H_0 \approx 68.3$ km/s/Mpc, 68% CL [67.6, 69.0], $> 7.2\sigma$ SH0ES exclusion via fix-M) that survived 14 adversarial challenges across 2 rounds, and caught 7 distinct error classes across 7 projects that pre-publication review missed.

The Hubble tension is a $> 7.2\sigma$ discrepancy in the Cepheid calibration of M , not the expansion history shape. The Roman Space Telescope (2027), Euclid, and Rubin will soon achieve ± 0.2 km/s precision and settle the question definitively.

We recommend adversarial validation as a standard step in data-driven astrophysics — cheap enough to be routine, systematic enough to catch errors humans and statistics miss, and reproducible enough to complement traditional peer review.

- 1 This research made use of PySR [Cranmer \(2023\)](#),
- 2 NumPy [Harris et al. \(2020\)](#), Astropy [Astropy \(2018\)](#),
- 3 and the Pantheon+ [Scolnic et al. \(2022\)](#), DES-
- 4 SN5YR [DES \(2024\)](#), and DESI [DESI \(2024\)](#) public data
- 5 releases. AI assistance (opencode) was used for code de-
- 6 velopment, adversarial review, and manuscript prepara-
- 7 tion. The complete analysis code and debate log will be
- 8 made public upon submission at [https://github.com/](https://github.com/ivan-hernandez/h0-symbolic-regression)
- 9 [ivan-hernandez/h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression).

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